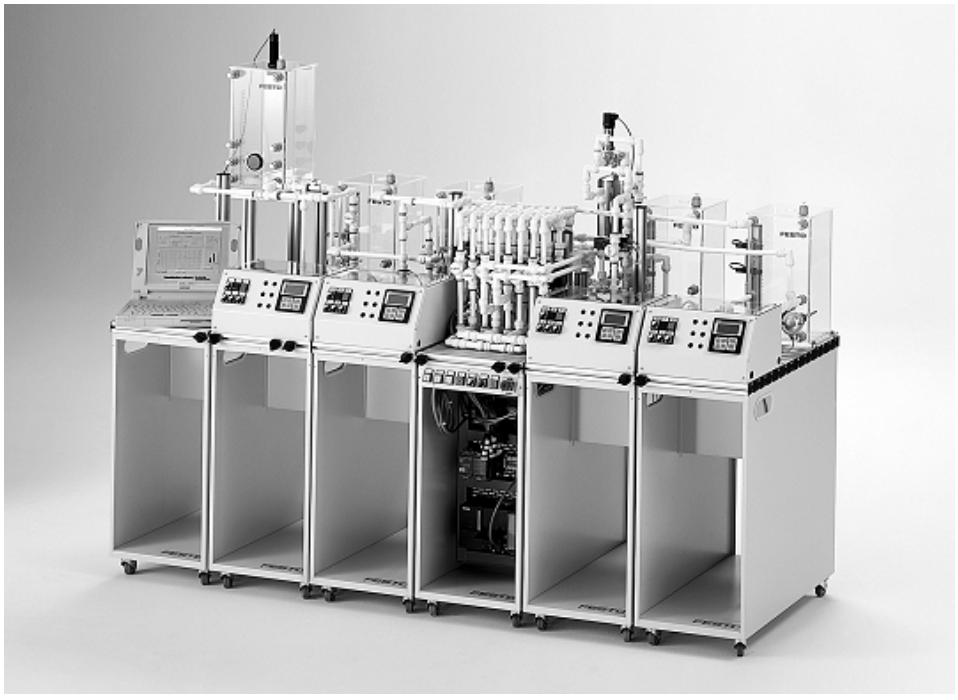
Process Control System

Control of temperature, flow, pressure and filling level

System Training – Complete System



FESTO

Published: 09/1999
Autor: Edward Gasper, Festo Singapore

© 1999 by Festo Didactic GmbH & Co.,
Rechbergstraße 3, D-73770 Denkendorf

All rights reserved, including translation rights. No part of this publication may be reproduced or transmitted in any form or by any means, electronic, mechanical, photocopying, or otherwise, without the prior written permission of Festo Didactic. Parts of this publication may be copied by authorised users solely for the purpose of instruction.

Table of contents

1 Training Schedule 5

2 Introduction 7
2.1 The Five Stations 8
2.2 Other Information 8
2.3 Control Panel 9

3 Installing Software 10
3.1 System Requirement 10
3.2 Software Provided 10
3.3 Installing Softing (Profibus) Software 10
3.4 Installing InTouch Software 11
3.5 Installing PCS Applications for InTouch 11
3.6 Configuration of Softing Profibus DDE Driver 11

4 Structure of the PCS System 12
4.1 Structure of the PCS System 12
4.2 Addresses, Dip Switches and Profibus Cable Connection 13

5 Fieldbus Technology 15
5.1 Fieldbus Technology 15
5.2 Profibus FMS - PCS Network Structure 15

6 Running the Process Control System 17
6.1 Starting the System 17
6.2 Controlling the Processes 19
6.3 Changing the Parameters 21

7 InTouch Software 24
7.1 Designing Using the InTouch Software 24
7.2 Opening a Window 25
7.3 Window Properties 25
7.4 Designing the Push Button for Level Control 26
7.5 Designing the Push Button for Flow Control 27
7.6 Designing the Visual Text Set Point 28
7.7 Designing Visual Process Value 29
7.8 Designing the Meter/Gauge Face to show the Process Value 30
7.9 Designing the Slider 32
7.10 Designing the Visual Level Tank. 33
7.11 Final Window 34
7.12 Testing the Window Design 34

8 DDE Communication 35
8.1 DDE Communication 35
8.2 Communication Between Software 36
8.3 Inserting a Dynamic Value in your Excel Spreadsheet 37
8.4 Inserting a Variable in InTouch 38

9 Using the PCS System as Single Stations 42
9.1 Changing the Controller Dip Switches 42
9.2 Changing the Controller Menu Settings 42
9.3 Installing EzDDE 42
9.4 Installing InTouch Applications 43
9.5 Starting the Individual System 44

10 Courseware 45

1 Training Schedule

Introduction
- Introduction to the Complete PCS System
- The Five Stations
- The Piping System
- The Control Panel

Installing Software
- Installing Profibus Card
- Installing Softing Software
- Installing InTouch Software
- Installing InTouch Applications for PCS

Structure of the PCS System
- Structure of PCS System
- Addresses and Dip Switches
- The Profibus Cables Connection
- The Bürkert Controller

Profibus Technology
- Fieldbus Technology
- Profibus FMS

Running the Process Control System
- Starting the System
- Controlling the Processes
- Monitoring the Processes
- Understanding the Software

InTouch Software
- Applications in InTouch
- Designing an Application Page

DDE Communication
- DDE Communication
- Communication between InTouch, MS Excel, Profibus Driver

Using the PCS System as Single Stations
- Changing the Controller Dip Switches
- Changing the Controller Menu Settings
- Installing EzDDE
- Installing InTouch Applications
- Using RS 232

Courseware
- Teaching of PID Control
- Courseware
- Presentation Foils
- Exercises

2 Introduction

The Process Control System may be used in initial vocational training, further training and retraining. The concept of the Process Control System is designed for professions such as process control electronics engineers, suppliers, process and waste disposal engineers.

With the stations of the Process Control System, the processes of filling level, temperature, pressure and flow control can be tested. The system consists of individual modules which can be combined in various ways.

Various degrees of expansion are possible in that stations and systems can be separately constructed and converted, piped up and commissioned. As such, they are open to your own ideas, approach to problem solving and project work.

Only components generally employed in industry are used in order to ensure a realistic training environment. This training concept is the ideal solution for process technology-related training compared to simulated systems and large industrial systems, which are confined to a particular area.

The following training contents can be taught by means of the Process Control System:
- Measuring of non-electrical, process technology and control technology variables.
- Extension of measuring chains into closed control loops.
- Assembly, interconnection and commissioning of systems.
- Expansion, modification, commissioning, inspection, maintenance and upkeep of process control equipment and error diagnosis in the event of malfunction.

The concept of the Process Control System is based on the following points:
- Modular system, step-by-step expansion possible.
- Practice-oriented thanks to the use of industrial components.
- Easy to transport system.
- The medium of water permits safe working.
- Open to new technologies such as process visualisation and process technology via Profibus.

The basic components of the Process Control System are:

- Containers
- Pumps
- Electrically actuated valves
- Manually actuated valves
- Heating unit
- Plastic push-in tubing
- Sensors
  - Flow sensors
  - Filling level sensors
  - Pressure sensors
  - Temperature sensors
- Industrial controllers
- Electrical components
- Mechanical components

2.1 The Five Stations

The Process Control System consists of five stations:

- Filing Level Station
- Flow Station
- By-pass Station
- Pressure Station
- Temperature Station

2.2 Other Information

- The tubing can withstand 28 °C at 12 bar and 90 °C at 2 bar.
- The material for the tubing is PEX.
- The temperature sensor is PT 100.
- The diameter for the tubing is 50 mm.
- PID controller used is Bürkert Controller.

2.3 Control Panel

At the By-pass station, there is a control panel, it is used for manual control of the By-pass station. This is used if you do not control the station with the InTouch software.

You need to switch the Auto/Manual button to Manual if you wish to control the By-pass station with the control panel.

- Start Button – To start the pump.
- Stop Button – To stop the pump.
- Second Button – To select which Input valve to control. It toggles between Level_In, Flow_In, Pressure_In and Temperature_In.
- Third Button – To select which Output valve to control. It toggles between Level_Out, Flow_Out, Pressure_Out and Temperature_Out.
- Quit Button – To reset the station.
- Emergency Stop – To stop the system.

3 Installing Software

For the Process Control System, there are a few software which needs to be installed.

3.1 System Requirement

- Pentium PC
- Minimum 32MB RAM
- Graphics VGA (640 x 480, 64K colours)
- Minimum 100MB Free Memory on Harddisk
- MS-DOS Version 5.0 of higher
- Windows 3.11 or Windows 95

3.2 Software Provided

- InTouch Development Program Disk (5 pieces)
- Softing GmbH Profibus FMS DDE-Server Version 5.0 (1 disk)
- PCS InTouch Application (2 disks)
- PCS Step 7 Files for S7, L, F, P, T (1 disk)
- PCS Profibus Files for DDE-Server Softing (1 disk)
- EzDDE DDE Driver (1 disk)
- PROVIS-GB Process Visualisation and Training (2 disks)

3.3 Installing Softing (Profibus) Software

- Before installing the software, ensure that the hardware card is placed in a vacant slot in the computer. You need to open the computer in order to insert the card.
- Connect the hardlock before installing the software.
- The software is from Softing. Read the manual before installing the software. There are some jumpers which might need switching.
- Remember to exclude the memory area as described in the Softing manual.
- In the Windows "System.ini" file include the entry "EMMEExclude=D000-D3FF" under the [386Enh] heading.
- Restart the PC to make sure the PC has excluded the memory area.
- Place the diskette in the disk drive.
- Go to Start and use the setup.exe file from the drive with the diskette with the Run program.
- Follow the directions given in the setup menu.
- Install the software as indicated in the manual.
- As there is only one Profibus card, you need to install the Profibus software only on one computer.

3.4 Installing InTouch Software

- The guide given below is just a very brief guide. You might need to read the manual given by InTouch to have a better understanding of the installation procedure.
- Connect the hardlock onto the computer's parallel port before installing the software.
- Place the diskette in the disk drive.
- Go to Start and use the setup.exe file from the drive with the diskette with the Run program.
- Follow the directions given in the setup menu.
- Upon completion of the installation, you need to reboot the computer to intialise it.
- After the installation of InTouch, you need to install the other application software.

3.5 Installing PCS Applications for InTouch

- Once the InTouch software is in, you can install the application software which is used with the PCS system.
- The software is given in diskette form and needs to be installed.
- Place the application diskette in the disk drive and run "setup.exe".
- Follow the directions given in the setup menu.
- The application files will be attached to your InTouch software.
- After completion, when you open Window Viewer, you should see the different available directories.
- Choose the application for the complete station. (C:\progra~1\pcs_s7pc).
- When the directory is highlighted, click on Window Viewer to see the Process Control System main screen.
- The other directories are for the complete system as well as for individual stations.

3.6 Configuration of Softing Profibus DDE Driver

All configurations of the driver are stored in two files; copy these files from the Profibus files disk (PCS Profibus Files for DDE Server Softing) to the folder where your Profibus driver is installed. You need to overwrite the two existing files.

The two files are :

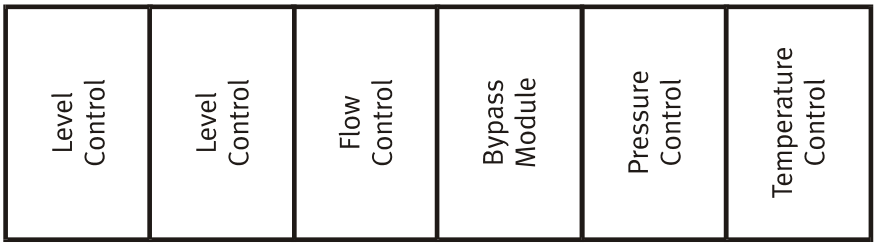
- Buspar.dat
- Kbl.dat

Only when the two files have been installed the Profibus driver would know the configuration of your PCS system.

4 Structure of the PCS System

4.1 Structure of the PCS System

- The complete PCS system consists of 5 modules:
    - Level Control
    - Flow Control
    - Bypass Module
    - Pressure Control
    - Temperature Control
- They are arranged as follows:

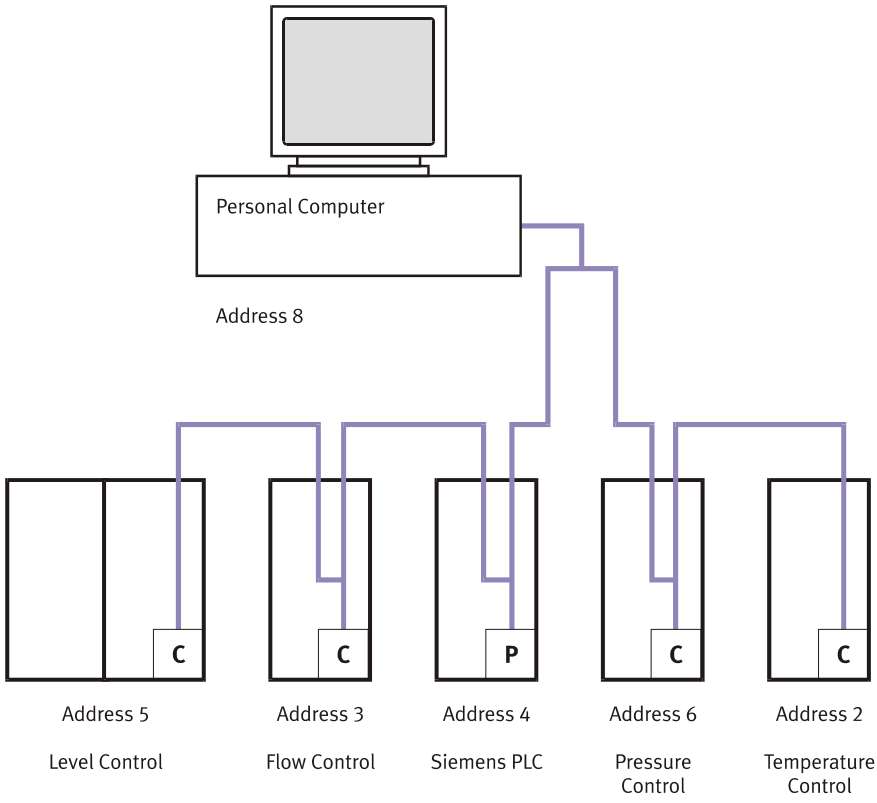


- This is only the physical arrangement of the different modules. If there are less modules, just remove them but try to maintain the order.
- If it is configured as a complete system, it will be linked through fieldbus. We will cover the fieldbus connections when we move onto the fieldbus section.
- The stations are also controlled individually by the Burkert controller. The manual will give a detailed flowchart on how to use the controller.
- It is advisable to try using the controller.
- Basically, when the controller is in the process mode it is running the system, you need to go to Configuration mode to do any changes to the parameters.
- At the end of the Configuration, you will go back to the Process mode.
- When you move from Process to Configuration mode, the output will remain at the last setting.
- When changing the settings for the controller, take note that there is no backward button, if you miss the setting, you need to go around once again to reach the same point.
- Keep track of where you are, this is helped by the flow chart in the manual.
- The recommended settings for each station is given in the PCS file (Page B7-4).

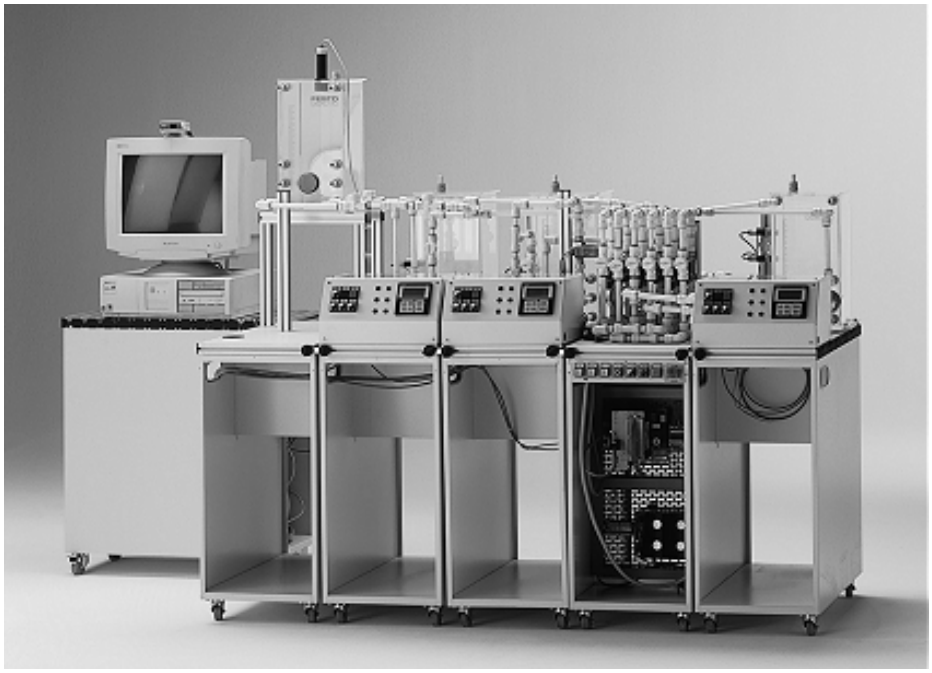
4.2 Addresses, Dip Switches and Profibus Cable Connection

When the Process Control System is used as a complete system as opposed to individual systems, follow the steps:

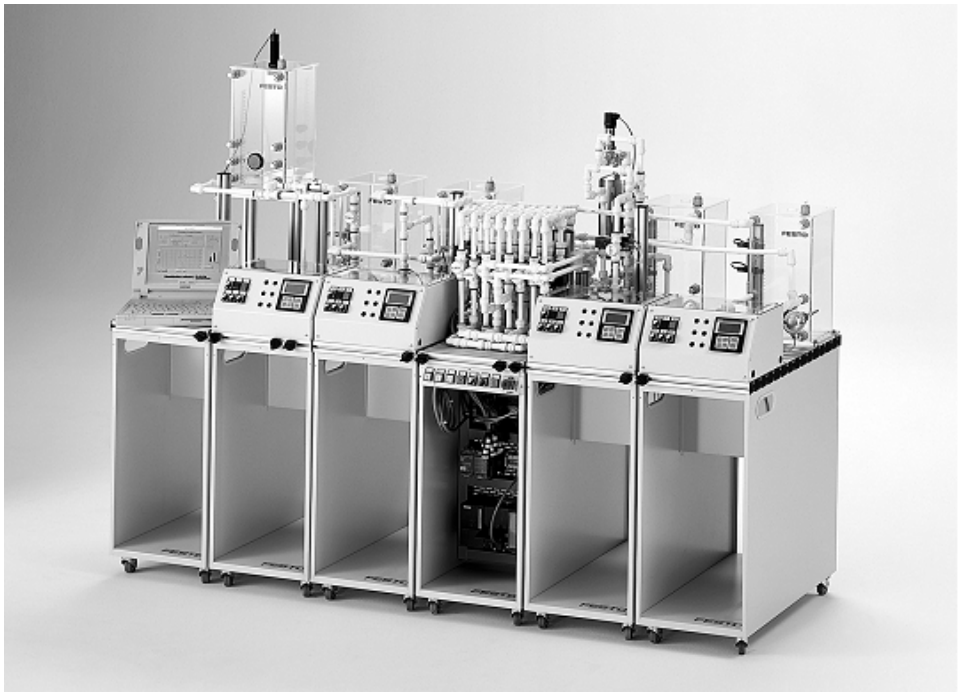
- Connect the Profibus cables as shown.
- Remove the cover of the control section to be able to access the controller.
- At the back of the controller, remove the screws and pull out the controller interface card.
- To use it as a complete station, the dip switches on the controller must be changed.
- The settings can be found in the controller manual.



- Set to Profibus setting.
- For the middle stations - Flow and Pressure
  (Off, Off, On, On, Off, Off, On, On)
- For the end stations - Level and Temperature
  (On, On, On, On, Off, Off, On, On)
- Reassemble the controller and replace the cover of the controller.
- Connect the Profibus cables as shown in the diagram. Remember to connect the short cable to the controller.
- Change the addresses of the controllers individually. This is done on the controller. Go to the Menu Serial and set the following:
  - Set to Remote.
  - Address for each controller. See B7-4 for table.
- Connect all the sensors to the Siemens PLC board.



PCS Complete System with 3 Stations



PCS Complete System with 4 Stations

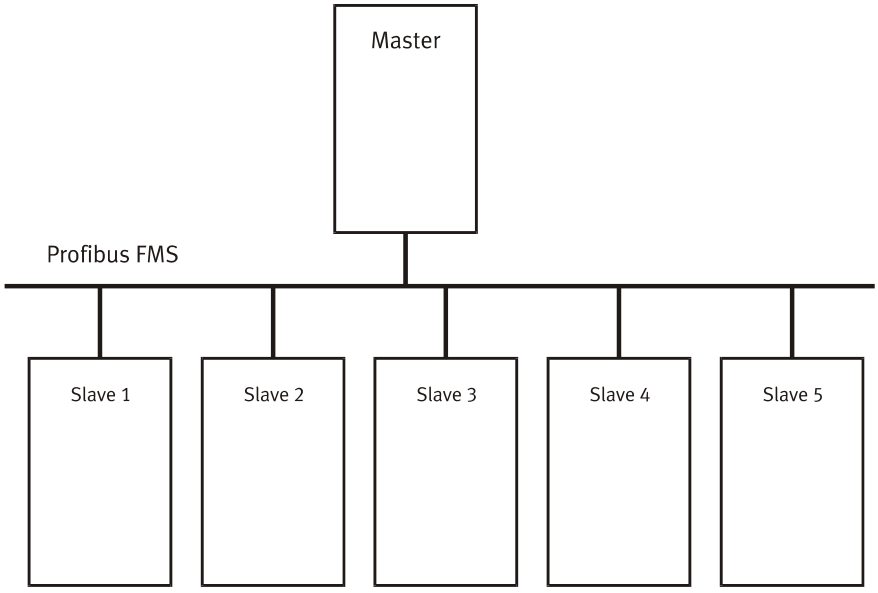
5 Fieldbus Technology

5.1 Fieldbus Technology

- The concept of fieldbus technology is found on the handout given.
- Fieldbus is an open communication system for industrial networking with a multi-vendor concept.
- As the PCS system uses Profibus, a powerpoint presentation on Profibus is attached.
- This presentation is from the Profibus association and can also be found on the web at the address : http://www.profibus.com
- There are three types of Profibus systems:
  - DP - Decentral Periphery
  - FMS - Field Message Specification
  - PA - Process Automation
- PCS uses Profibus FMS.

5.2 Profibus FMS - PCS Network Structure

- The basic Profibus structure is shown below:



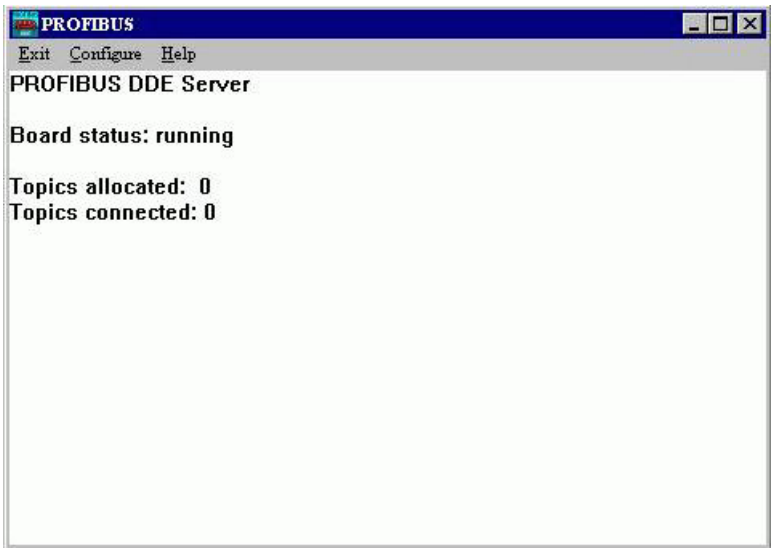
- In the Master there are two important files:
  - Communication Reference List (KBL, CRL)
  - Bus Parameter Settings (Buspar.dat)

- In the Slave there are:
  - Slave Address
  - Object List - Index, SubIndex, Type
- To open the KBL or Buspar.dat, use only Notepad and not any other word processing programs as they are written in computer language and any formatting will affect the program.
- Your KBL must relate to your hardware settings.
- You can also do the changes in the Profibus software, go to Configuration.
- You should be able to get the parameters for the files from the controller manual and also from the Profibus driver manual.
- If there are more than one master, there is time sharing between the masters by token ring.

6 Running the Process Control System

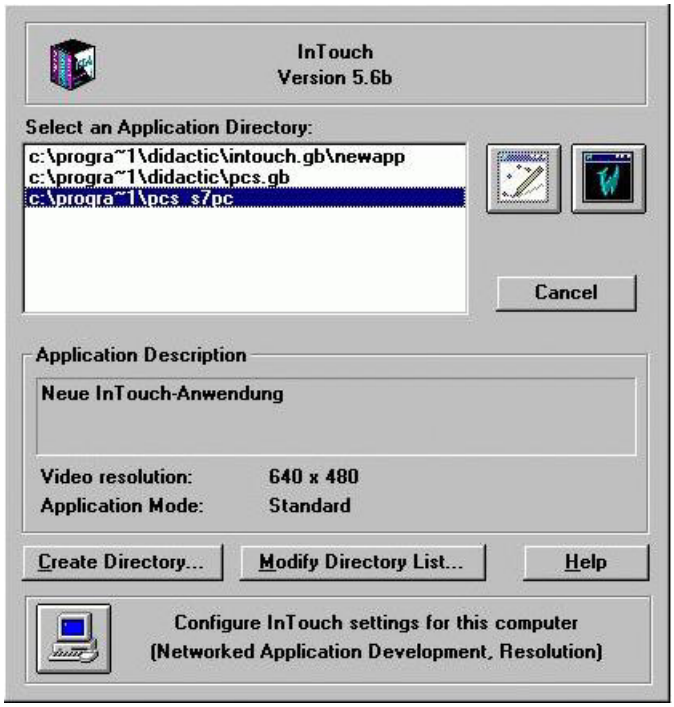
6.1 Starting the System

- Switch on the complete system.
- Make sure that the "On" switch for each individual station is switched on.
- Press the "Emergency Stop" button on the By-pass station and pull it out. Press the "Quit" button to reset the system. The light on the Emergency button should go off.
- Start the computer with all the installed software, make sure the hardlocks are connected.
- Start the Profibus driver first, go to "Start", "Programs", "FMS-DDE" and "FMS-DDE Server".
- When the dialog box appears, make sure that the "Board Status" is "Running" and not "Error".
- If it shows "Error" restart the Windows program, it should be running after that.

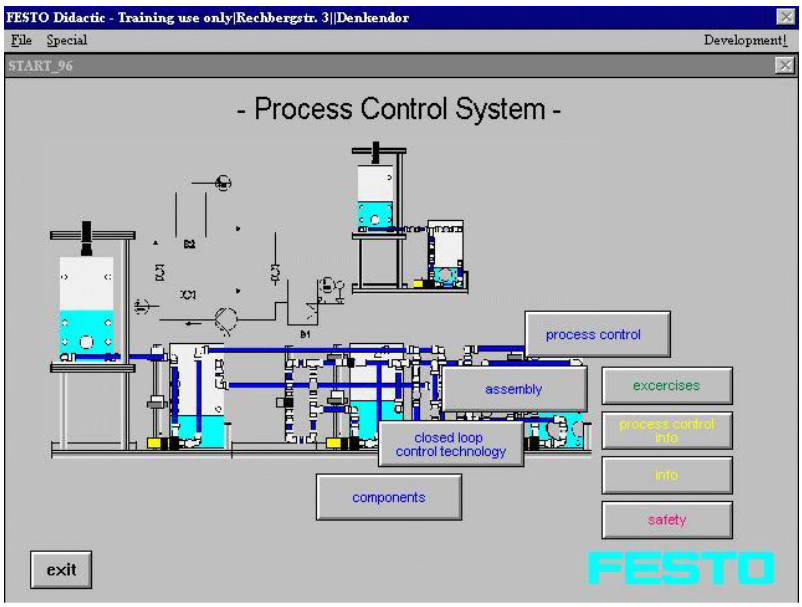


- Minimise the Profibus window.
- Start InTouch programme. Go to "Start", "Programs", Festo Didactic" and "InTouch".

- The InTouch screen would appear, make sure the correct directory is highlighted (c:\progra~1\pcs_s7pc).

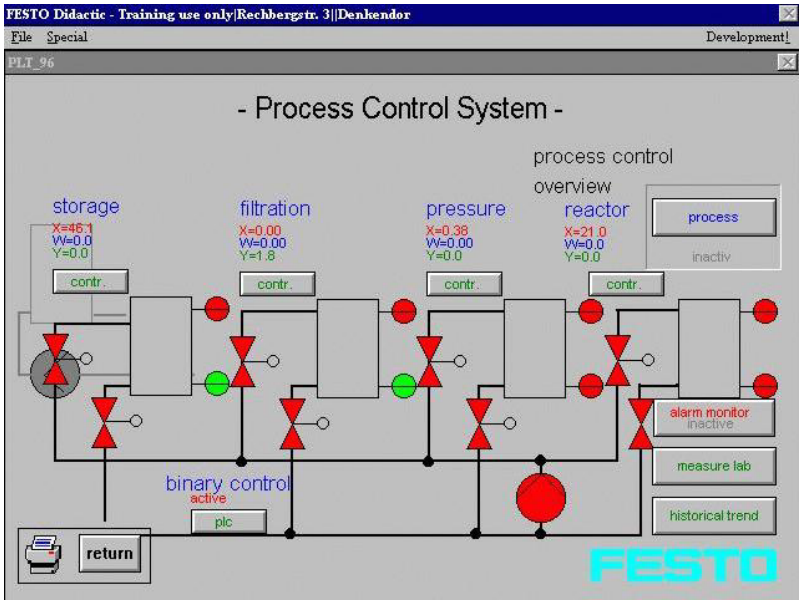


- Click on the "Window Viewer" icon to launch the application.
- You should see the Process Control System screen.

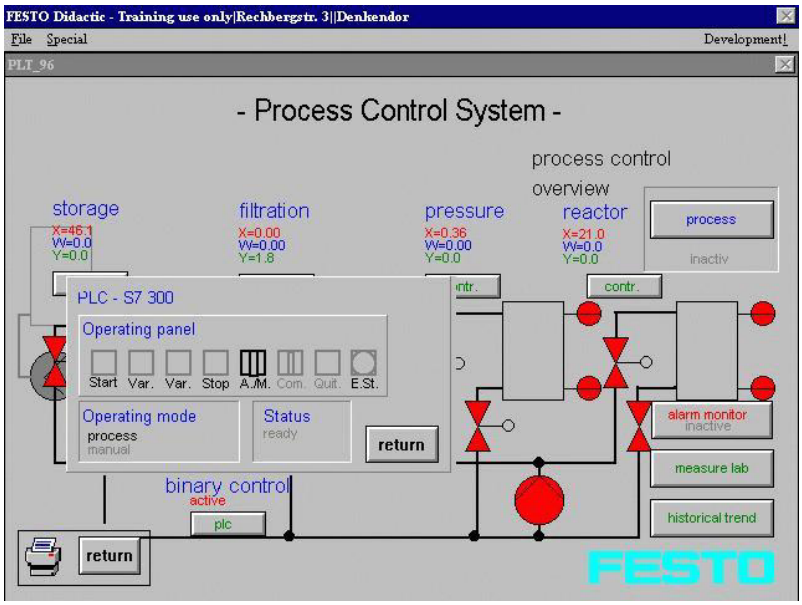


6.2 Controlling the Processes

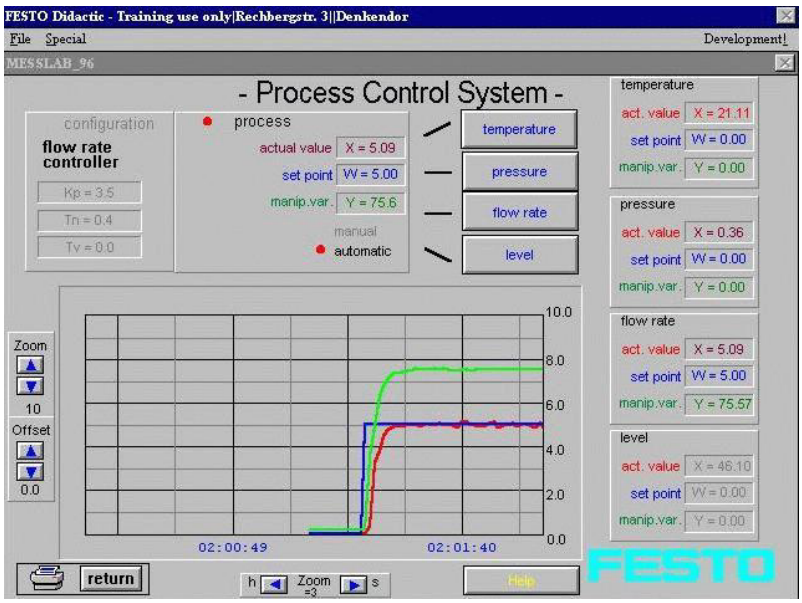
- Click on Process Control.
- You should see the By-pass station with all the valves and sensors.



- Make sure that the PLC is active.



- You can control all the valves and pump from this screen.
- If it is in green colour, it means that is open or for the pump, switched on.
- Click on "Measure Lab" to go to the control screen for all the stations.



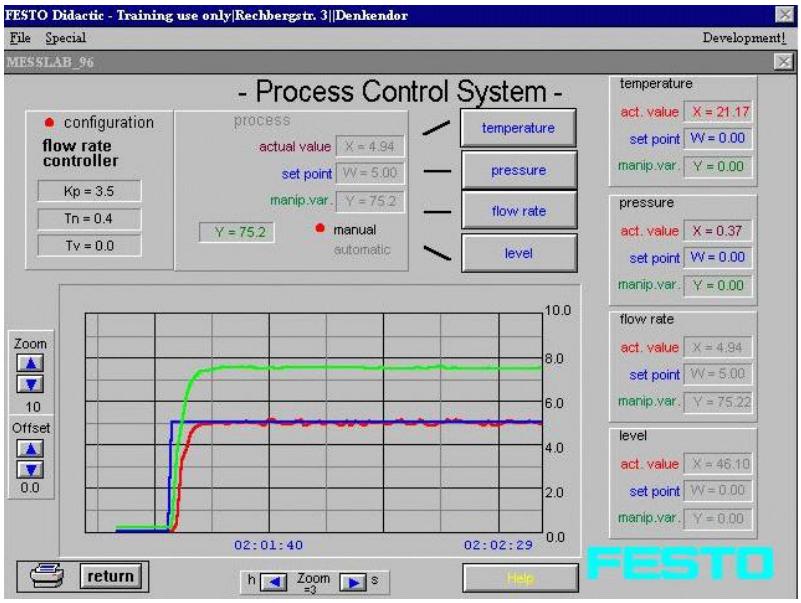
- You can choose to control the different stations in this screen.
- As it is a complete system, you are able to control more than one station simultaneously.
- However, your measure lab graph can only display one process at a time.
- You can get the standard settings for the different processes in your manual (Page B7-4). However, some of the important data are:

	Level	Flow	Pressure	Temperature
Kp	28	3.5	0.5	4
Tn	5	0.4	0.5	2500
Tv	0	0	0	0

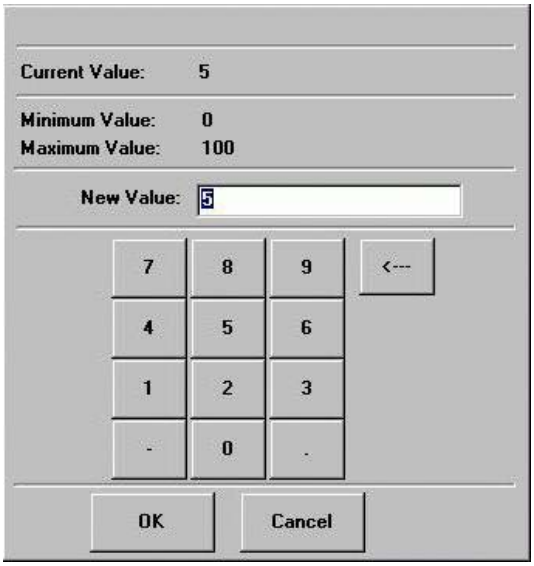
- The measure lab is the tool especially designed for exercises I closed loop control technology. You can monitor step-responses of each sub-system - adjust, analyze and optimize the controllers parameters.

6.3 Changing the Parameters

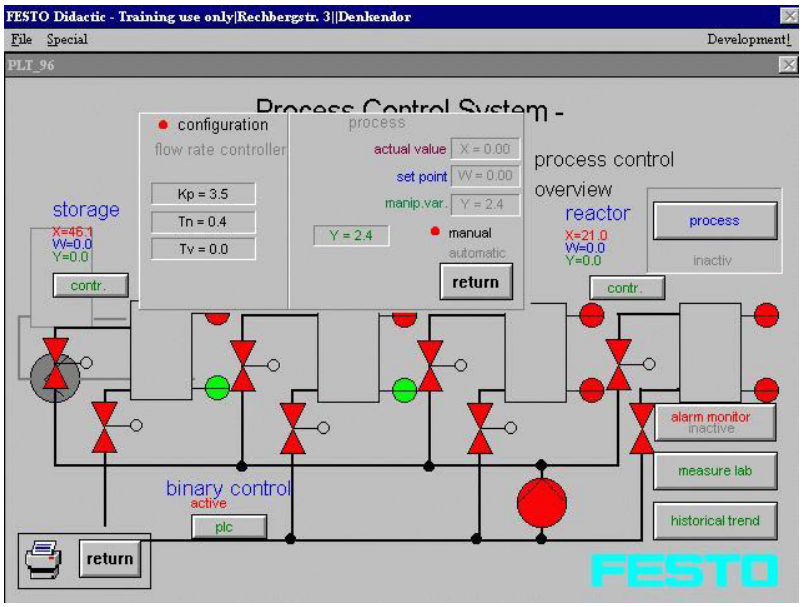
- Choose the process you wish to control by clicking on the relevant button.



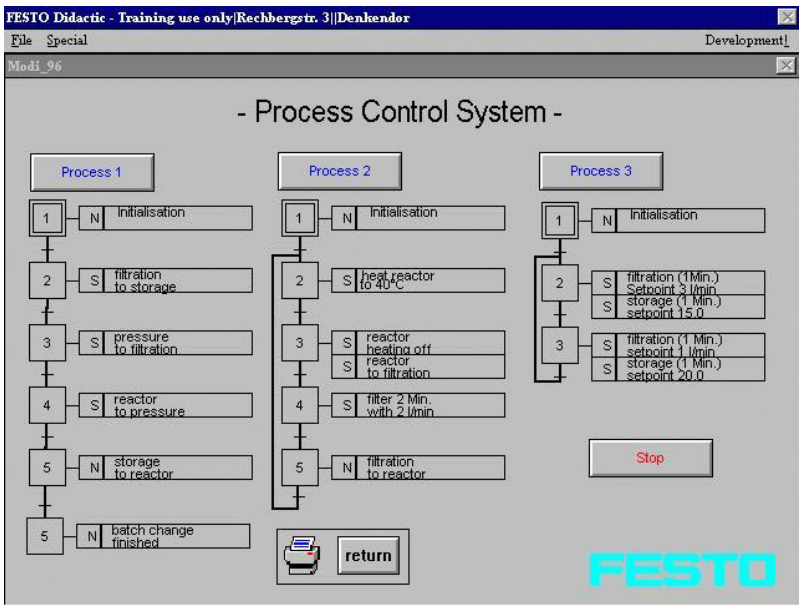
- After choosing the process, click on "Configuration" to change the parameters; Kp, Tv and Tn.
- After changing the parameters, click on "Process". Here you can change the "Set point".
- Make sure that you are in manual mode before changing the set points.
- Click on "Set point". A dialog box similar to a calculator will appear. Enter a new set point and click "OK".



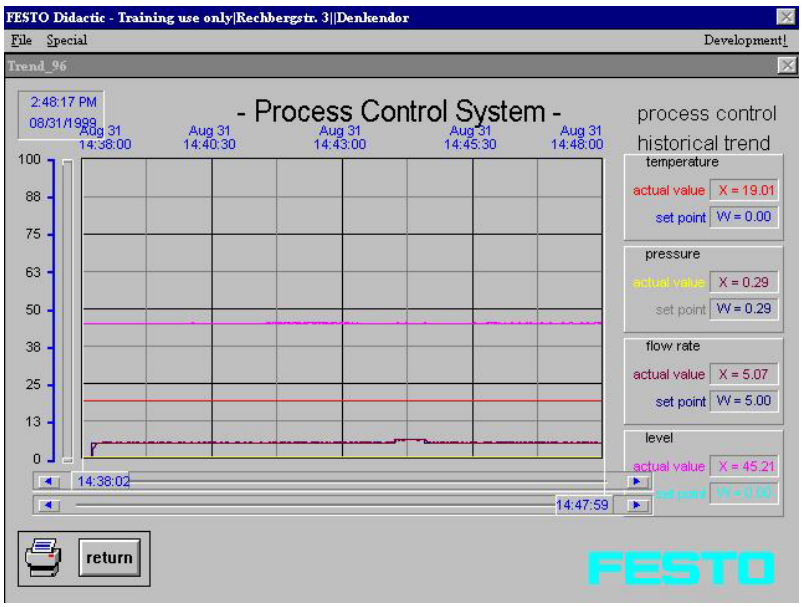
- Click on "Automatic" to set the controller. The LED of the Bürkert controller will go on.
- The controller will now proceed to meet your set point.
- You can also change the parameters when you are in the screen with the by-pass station.



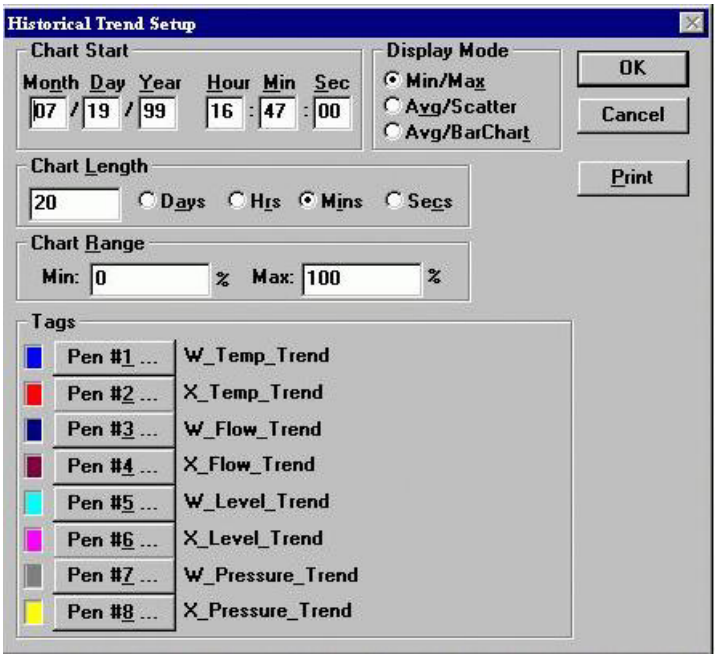
- If you click on the "Process Control System", it will go to process mode. Refer to D1-2 for more information.
- The following screen will appear.



- If you click on "Historical Trend" you will be brought to this screen.
- It will show the readings for the past specified period.
- You can check the documentation D1-6/7 for more information.



- If you click on the Data Screen, a Historical Trend Setup screen will appear.
- You need to change the settings for the "Period" (Chart Start) and "Monitoring Time" (Chart Length).



7 InTouch Software

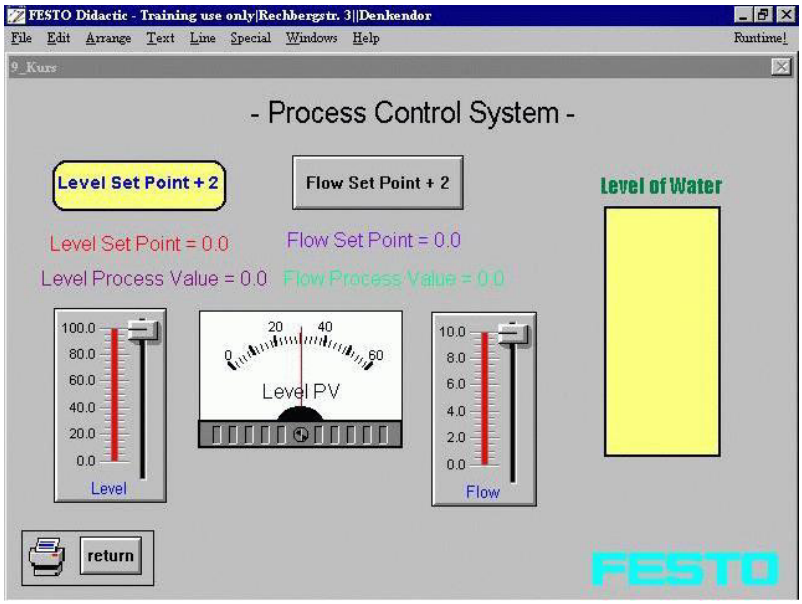
The InTouch software is an easy way to create man-machine interface (MMI) applications for the Windows operating system. The InTouch software consists of two major components: Window Maker and Window Viewer, and six utility/diagnostic programs.

Window Maker is the development environment, where object-oriented graphics are used to create animated, touch-sensitive display windows. These display windows can be connected to industrial I/O systems and other Windows applications.

Window Viewer is the runtime environment used to display the graphic windows created in Window Maker.

7.1 Designing Using the InTouch Software

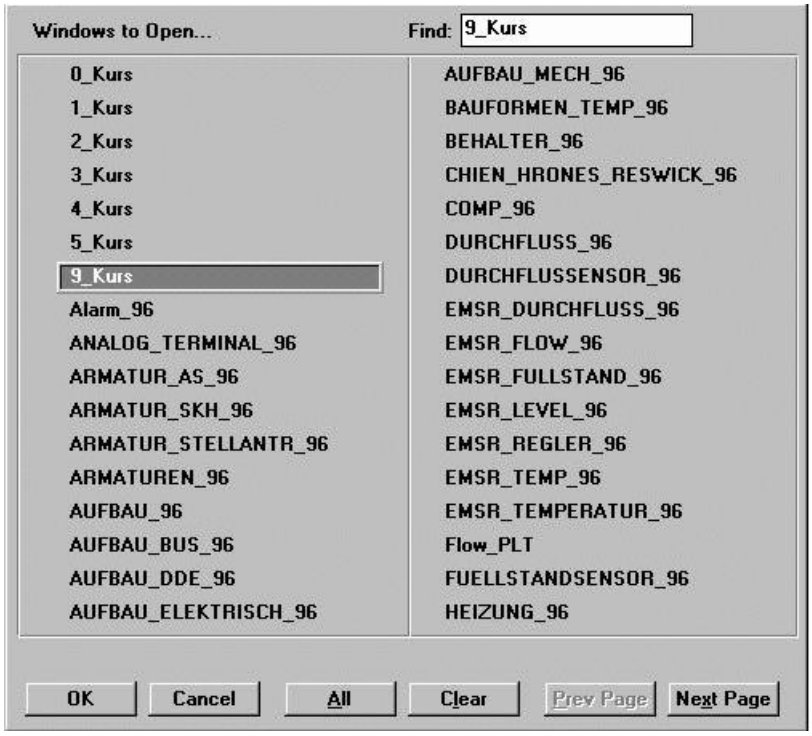
- To be able to understand the InTouch software, we will be designing a simple page using some of the features.
- We will try to design the following page.



- In this exercise, we will try to design the page with the following:
  - Buttons, when pressed will increase the setpoint by a value of 2 for both the Level and Flow control separately.
  - Text boxes which will show the setpoint and process value.
  - A meter to indicate the Process value for Level control.
  - Sliders to change the setpoint for Level and Flow control.
  - A level tank to indicate the level of liquid.

7.2 Opening a Window

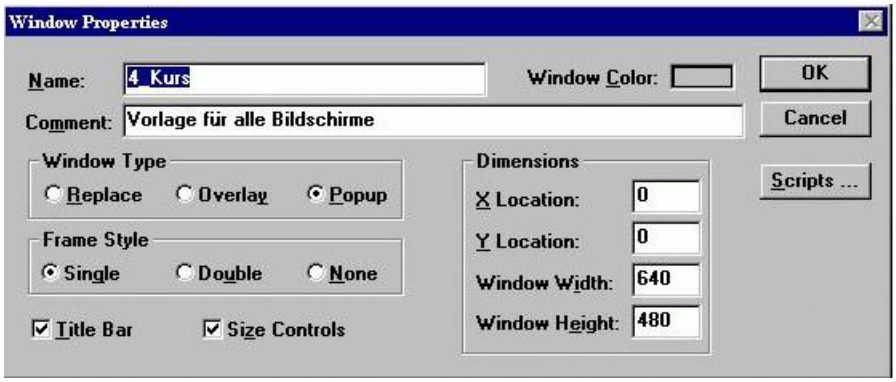
- To design the above, you must go to InTouch Window Maker.
  - Open 4_Kurs file to do the above.



  - Highlight and delete the present features on the window.
  - This file is related to the Custom Exercises on the main PCS window.

7.3 Window Properties

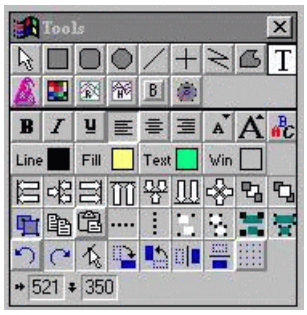
- To change the properties of the Window, go to Windows and then Window Properties.



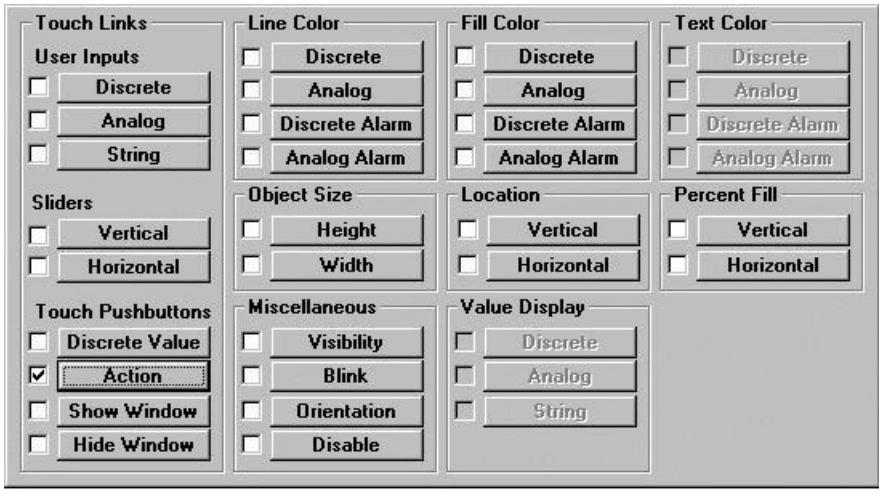
- Please refer to the InTouch manual for more details on the options.

7.4 Designing the Push Button for Level Control

- Designing the Push Button to Increase the Set Point of the Level Process by 2.
  - Create a push button by using the "Rounded Rectangle" icon from the Draw Object toolbar.

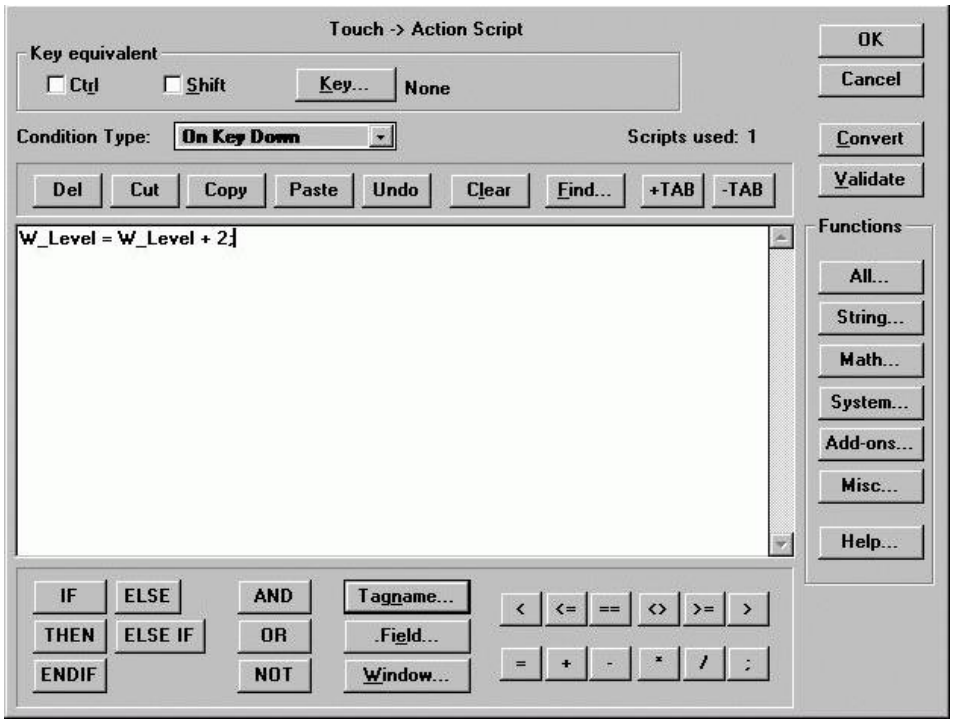


  - Click on the "Text" icon to write the text on the button.
  - Type "Level Set Point + 2"
  - This is to indicate what the button is for.
  - You can change the font by clicking on the Font icon.
  - The default font is small fonts which is why you cannot change the size of the font by clicking on "Enlarge Font" icon.
  - You can change the appearance of the text by using the different icons on the toolbar.
  - After you have finished typing the text, you cannot change the text by double-clicking on it. You will have to press [Ctrl] + [L] to bring you to the text properties dialog box.
  - You can change the text here.
  - Double click on the button to get into the Dialog box.
  - Mark on the Push Button Action.



  - Double click on the Action button.

- In the Action script, enter the expression "W_Level = W_Level + 2;"



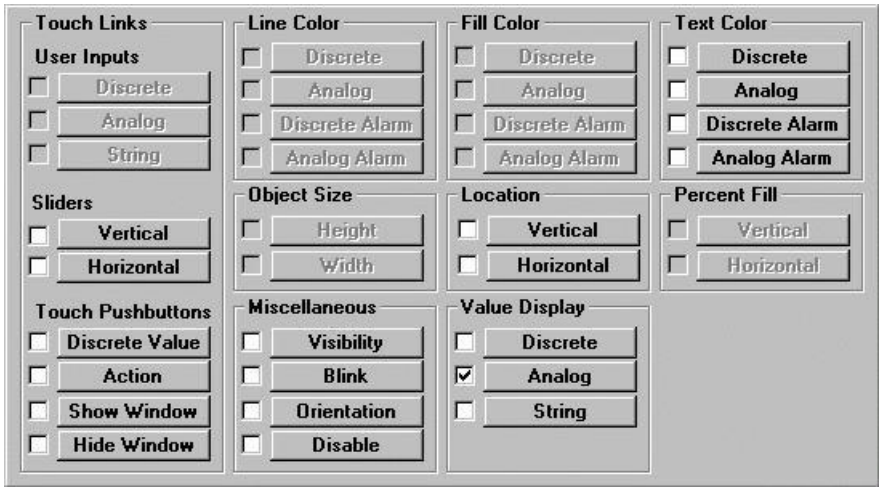
- Click on OK to return to previous menu and click on OK again.

7.5 Designing the Push Button for Flow Control

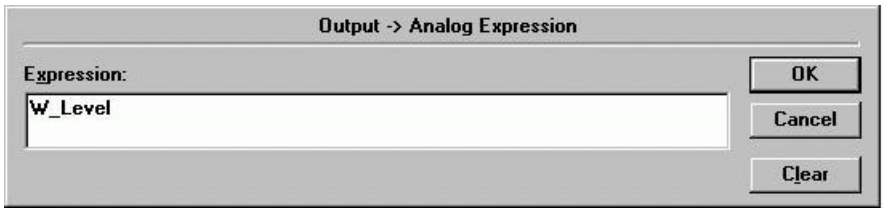
- Designing the Push Button to Increase the Set Point of the Flow Control by 2.
  - Create a push button by using the "Buttons" icon from the Draw Object toolbar.
  - Similar to designing the Level button, only this time the button is pre-designed and with text.
  - To change the text press [Ctrl] + [L].
  - Double click on the button to get into the Dialog box.
  - Mark on the Push Button Action.
  - Double click on the Action button.
  - In the Action script, enter the expression "W_Flow = W_Flow + 2;"
  - Click on OK to return to previous menu and click on OK again.

7.6 Designing the Visual Text Set Point

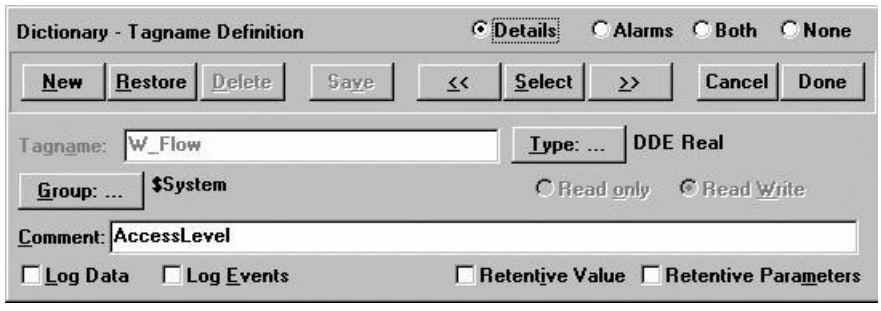
- Designing the Visual Text Set Point.
  - This text is designed to show the current value of the set point.
  - Click on Text from the Drawing toolbar and type in the following : "Set Point = 0.0". The number will indicate the number of decimal places you wish to see.
  - Once completed click elsewhere on the screen and then double click on the text again. This will bring you to the dialog screen.



- Mark Value Display Analog and click.
- This will bring you to the "Output → Analog Expression" box.



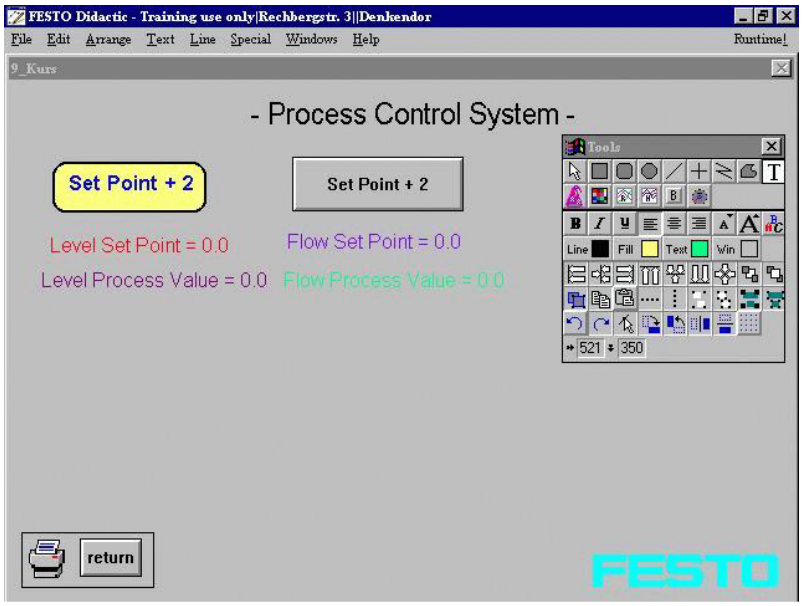
- Double click on Expression and you will see the Tagname Dictionary box.



- Enter "W_Level" as tagname and I/O Real as type.
- Exit by closing the boxes.
- Repeat the same procedure for Flow control.

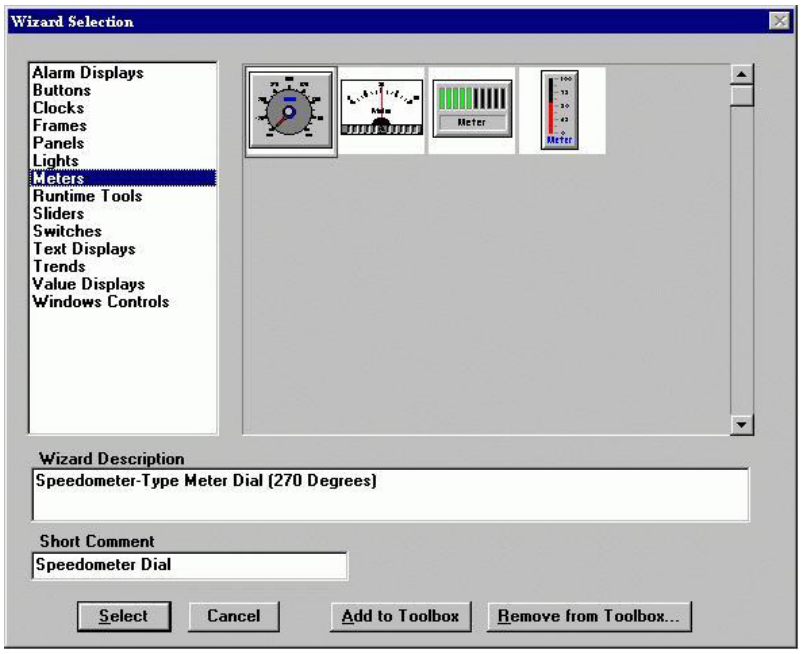
7.7 Designing Visual Process Value

- Designing the Visual Process Value.
  - It is similar to the Visual Set point settings except that you must choose "X_Level" and "X_Flow".
  - By now, your screen should look like this:

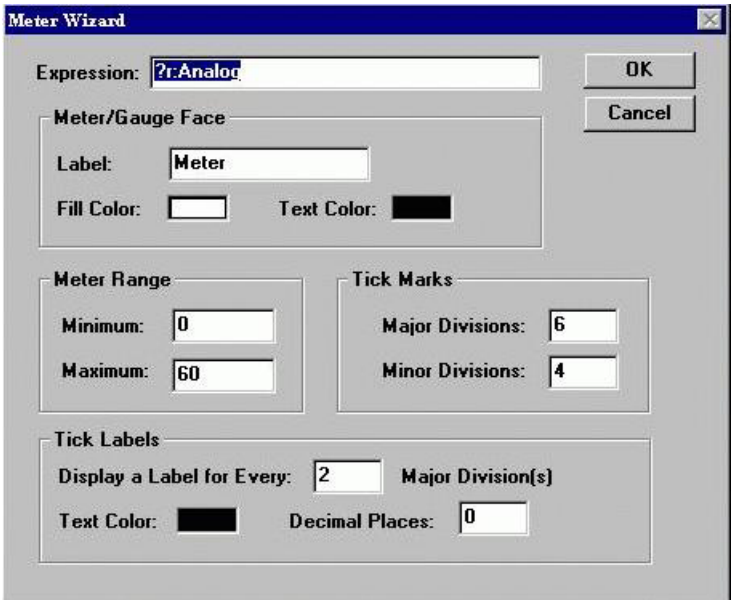


7.8 Designing the Meter/Gauge Face to show the Process Value

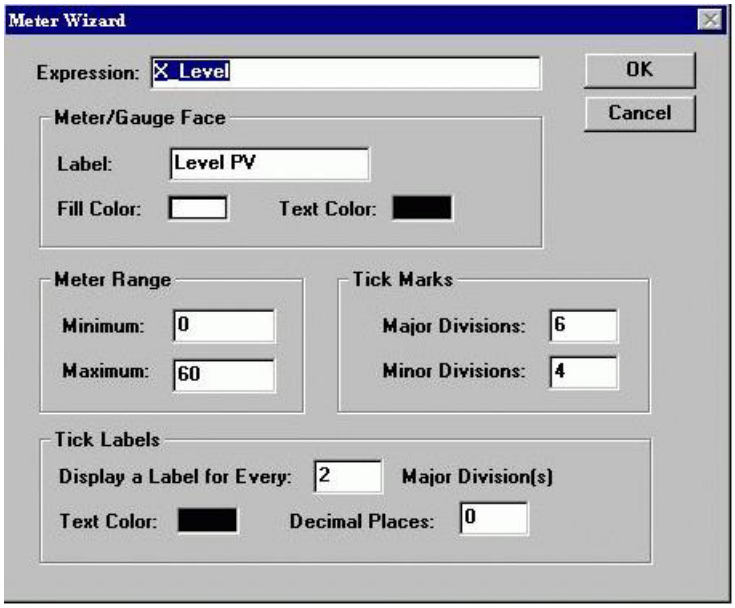
- Designing the Meter/Gauge Face to show the Process Value.
  - Click on the Wizards icon on the main toolbar.
  - Select Meters on the left box and choose the gauge you want to use in the right box.



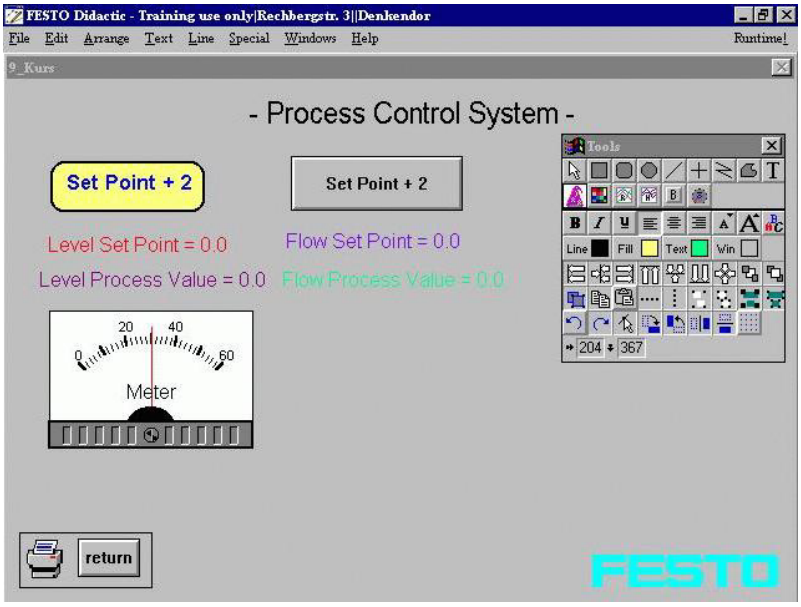
  - Double click on it and you will return to your workspace. Move your cursor to the location you wish to place the meter and click.
  - The meter will appear.
  - Double click on the meter and the Meter Wizard box will appear.



- Double click on expressions and the Tagname Dictionary will appear. Choose "X_Level" and DDE Real type.

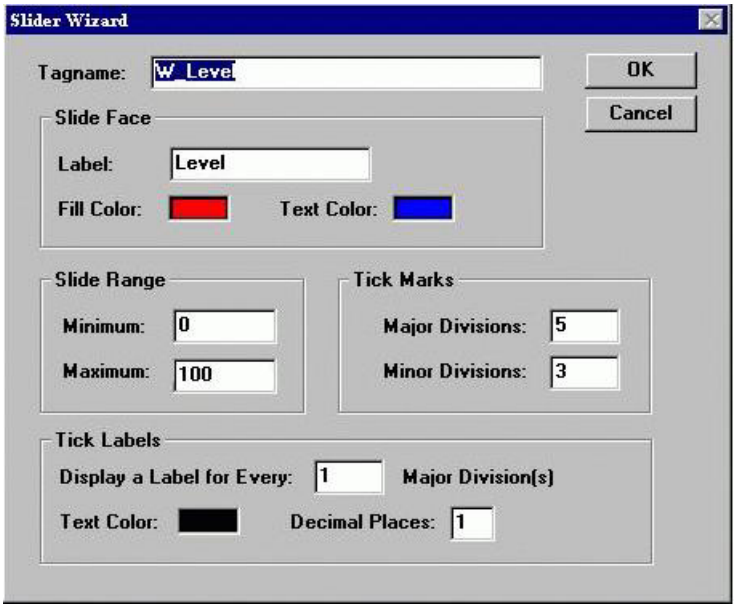


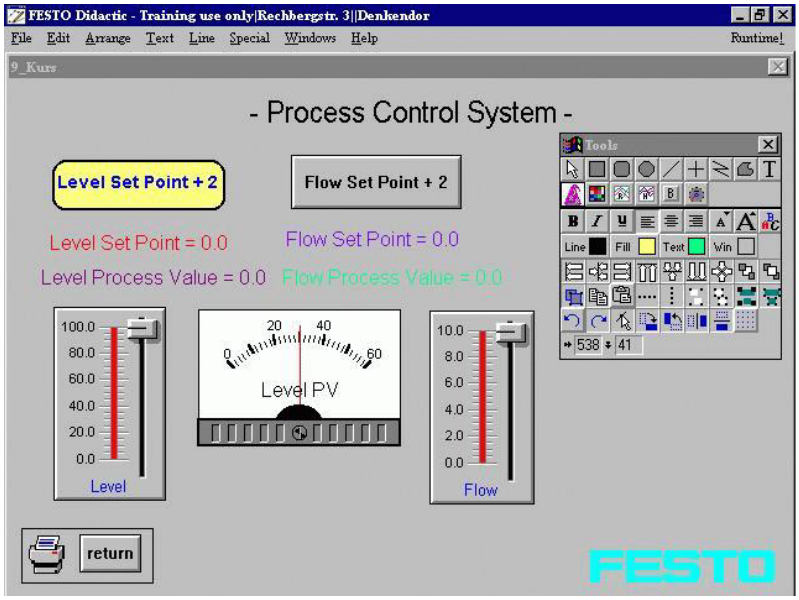
- Click on Close and you will return to the previous screen. Change the label to "Level" and click OK.
- Repeat the same procedure for designing a gauge for Flow.



7.9 Designing the Slider

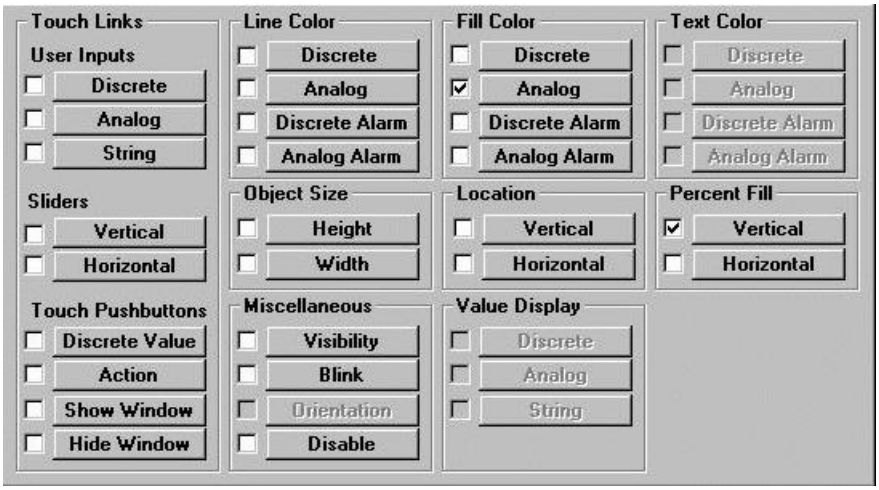
- Designing the Slider.
  - Similar to designing the meter but choose Sliders and change the tagname to "W_Level".
  - You may wish to change the "Slide Range" and "Tick Marks".
  - You can also add text on top of the slider to indicate what it is used for.

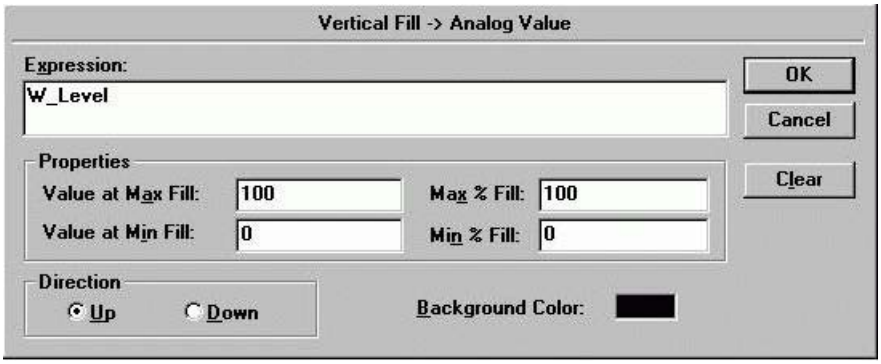




7.10 Designing the Visual Level Tank.

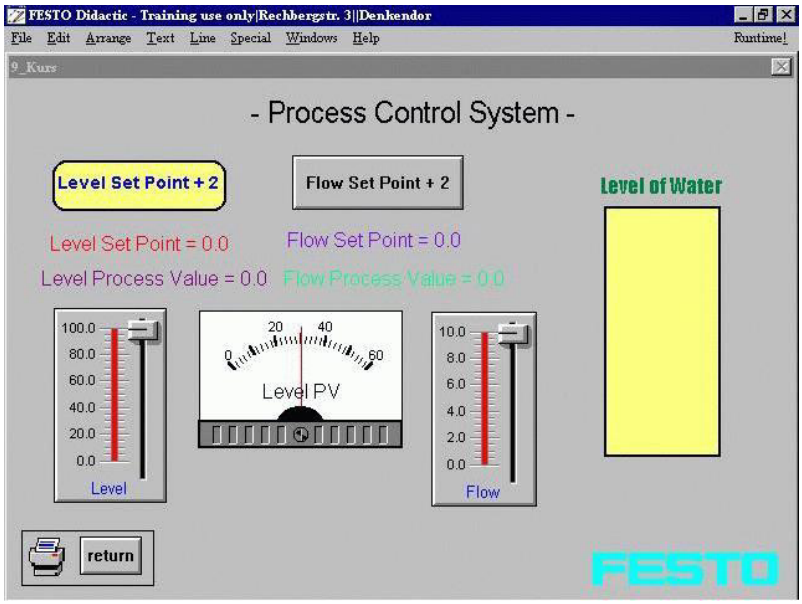
- Designing the Visual Level Tank.
  - Create a tank with the "Rectangle" icon from the drawing toolbar.
  - Double click on the rectangle to bring you to the Dialog box.
  - Choose Fill Color Analog and Percent Fill Vertical.
  - Double click to enter the Expression W_Level.
  - You can also add text to indicate what it is showing.





7.11 Final Window

- Your final window should look like this.



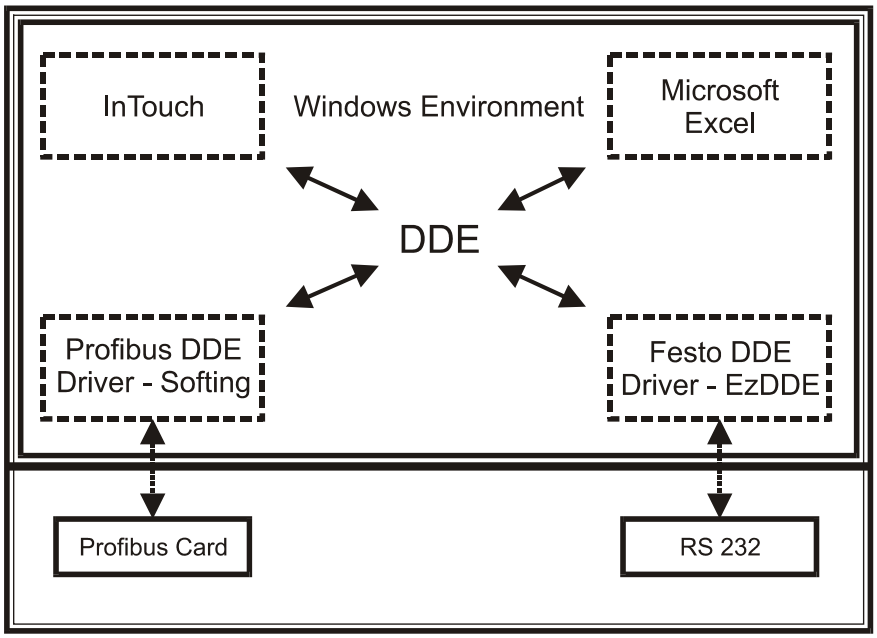
7.12 Testing the Window Design

- To see if your design works, exit and go to Windows Viewer to check.
- Make sure that your system is connected to have communication between the computer and the PCS system.
- Also ensure that the Level and Flow control are on automatic mode so as to enable you to manipulate the figures.

8 DDE Communication

8.1 DDE Communication

- DDE stands for Dynamic Data Exchange and can be found in most Windows programmes.
- DDE is used for the communication between the different software programmes.
- For the PCS system, the structure is shown below:



- When we use DDE communication, three things are important:
  - Application
  - Topic
  - Item
- For each of the application, there are different codes which can be found in the software manual.
- To find out more about DDE, you can refer to the manual which comes together with the software.

8.2 Communication Between Software

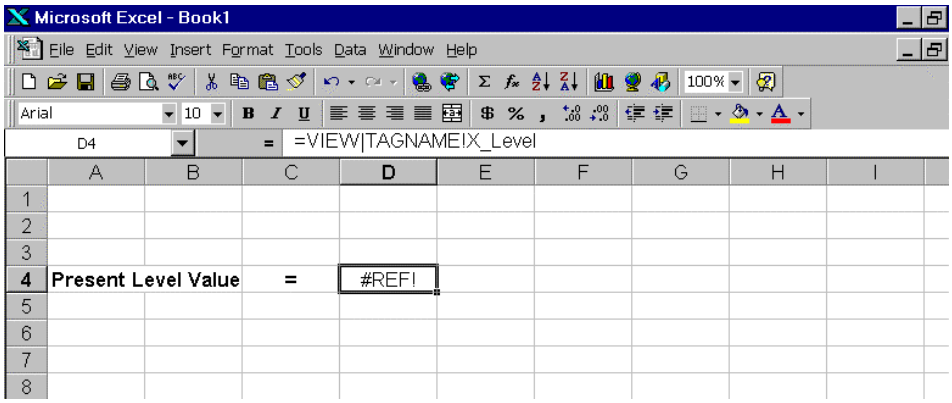
- Before communicating between the different software, we must know the codes.
- The following are the codes for the software that we are using:

	MS- Excel	InTouch	EzDDE	Softing DDE Server
Application	EXCEL	VIEW	EZDDE	PROFIBUS
Topic	Sheet Name eg. TEXT1.XLS	TAGNAME	REGLER	Defined in Topic Definition Box Eg. KR2 KR3
Item	Row & Column eg. R3C2	eg. X1 W1 X_TEMP	eg. X1 W1 K_P	Defined in the Manual eg. O21T8I500

- For the Softing DDE Server; in the item category:
  - O - Object
  - T - Data Type
  - I - Calling Interval
- By knowing the above, you can actually place dynamic values in any of the above programmes.
- For example, you can see the process value of the flow control station by inserting a DDE into your Excel spreadsheet.

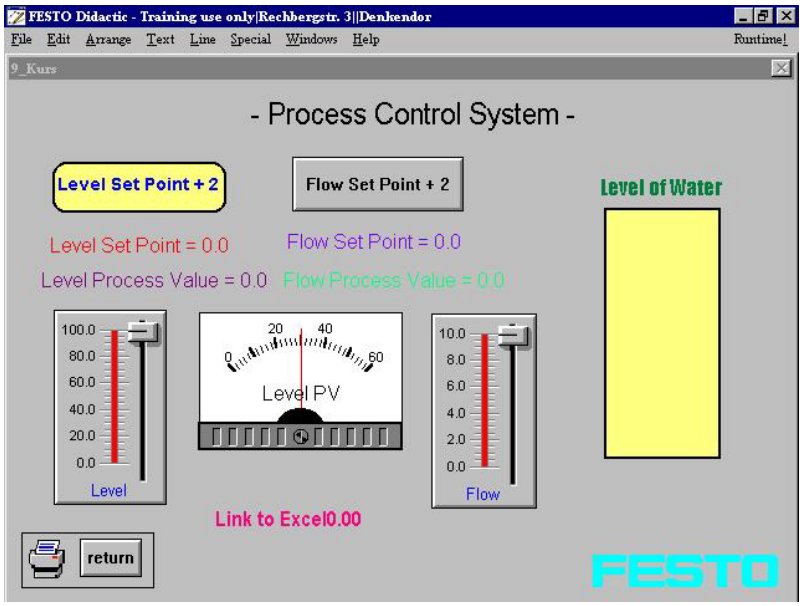
8.3 Inserting a Dynamic Value in your Excel Spreadsheet

- Inserting a dynamic value into your Excel spreadsheet.
  - The link is created by typing "=Application|Topic!Item"
  - The Application and Topic is separated by a | and the Topic and Item by an exclamation mark.
  - For example, go to a cell in Excel and type "=VIEW|TAGNAME!X_Level" and you would get
    the process value as it changes.

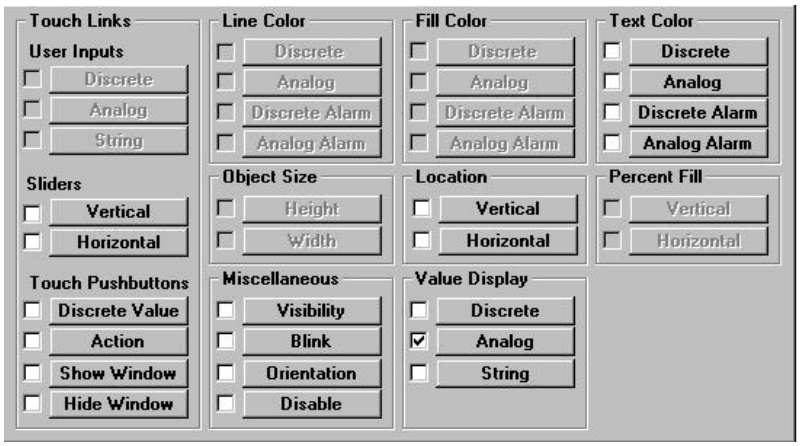


8.4 Inserting a Variable in InTouch

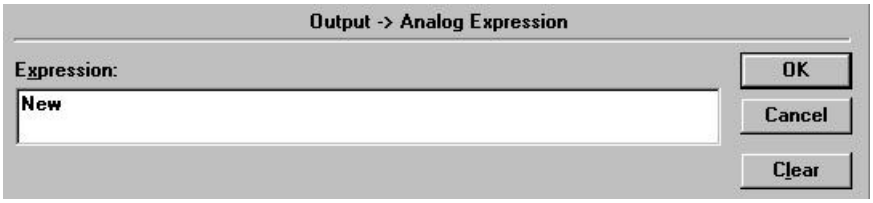
- Inserting a variable into InTouch.
  - Open InTouch Window Maker.
  - Open Window 4_Kurs or 9_Kurs.
  - Create a variable on any available space, for example, type "Link to Excel 0.00"
  - Click on the open workspace and then double click onto the text.



- It will bring you to the dialog box.
- Mark on Value Display - Analog.
- Double click on Analog and you will be brought to the „Output → Analog Expressions“ box.



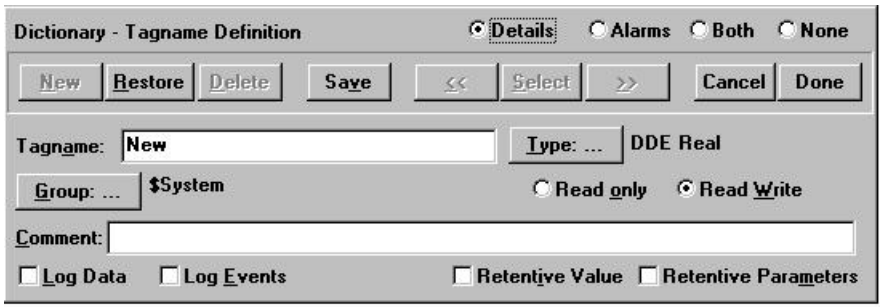
- In the Expressions, type in a name, e.g. New.
- Click „OK“.



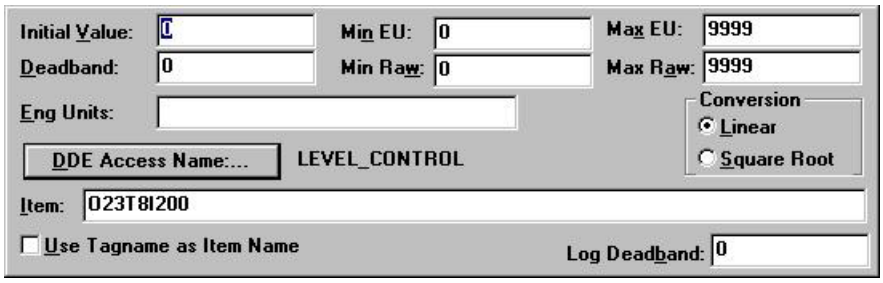
- Click on „OK“ when asked to define „New“.



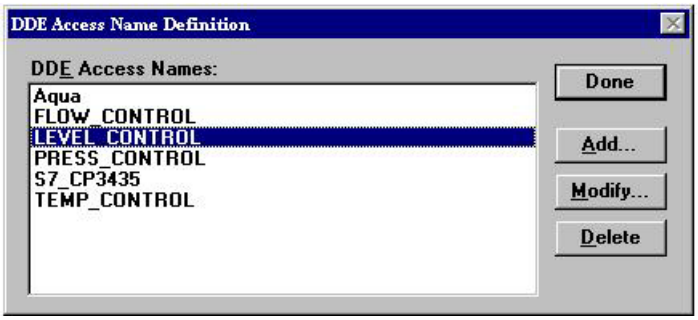
- Click on „Type“ to change to DDE Real.



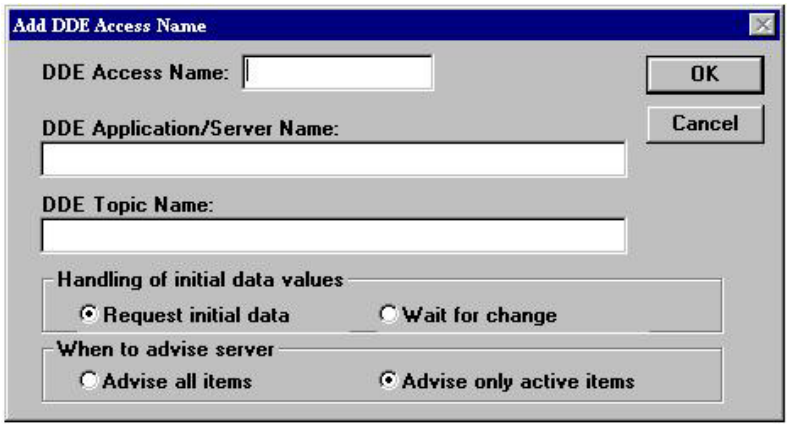
- Click on „DDE Access Name“.



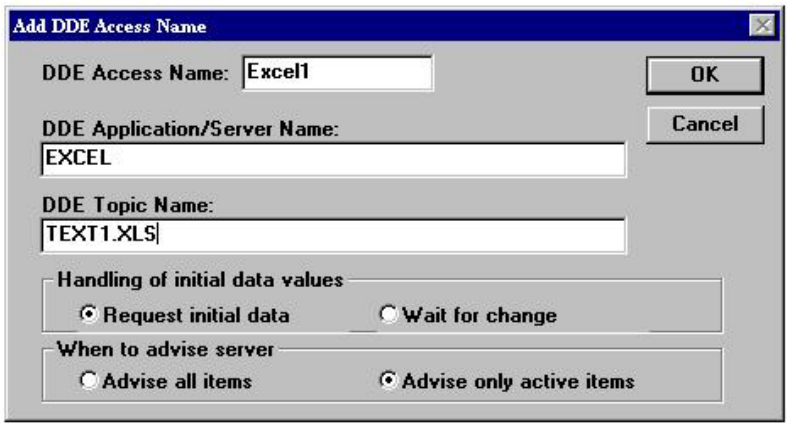
- Click on „Add“.



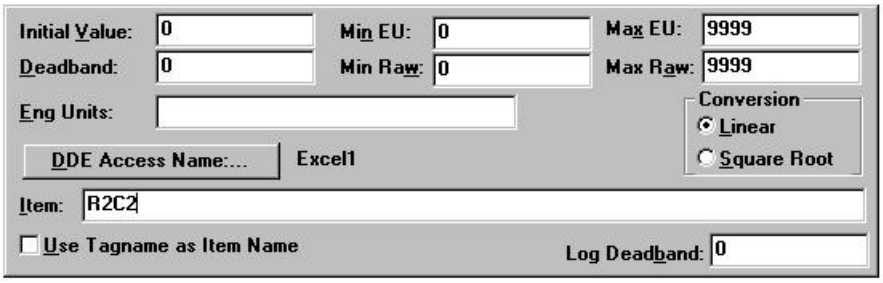
- You will see „Add DDE Access Name“.



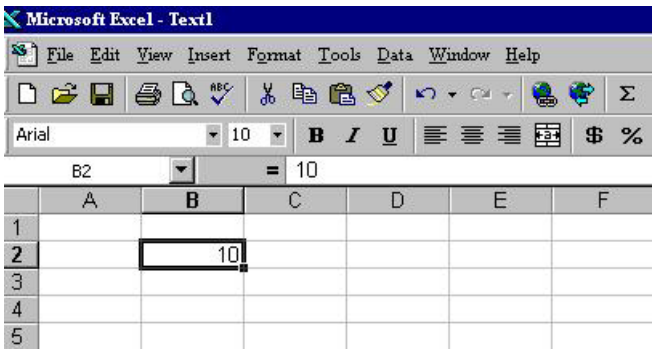
- You can give it any Access Name but you must type in the right links for the Application and Topic Names.
- To link to Excel, type "EXCEL" for Application Name and "TEXT1.XLS" for Topic Name. Click Ok and Close.



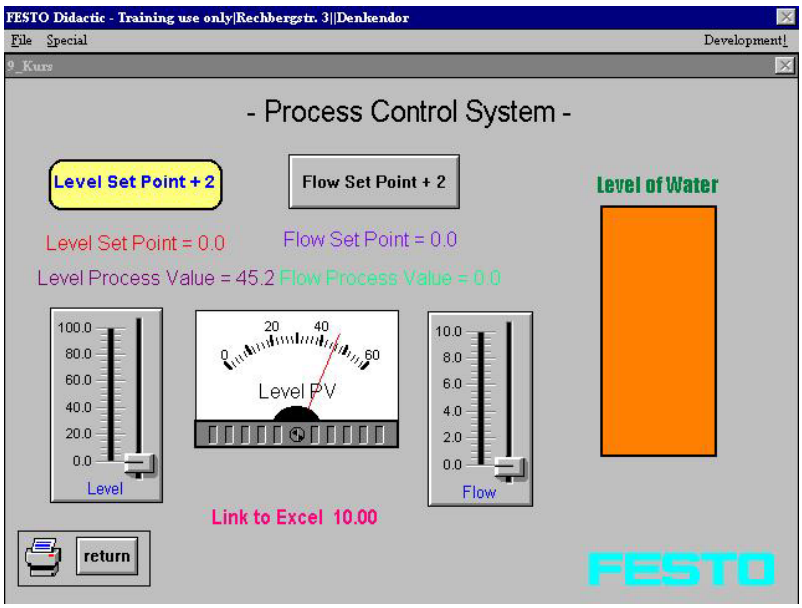
- Back to the Tagname Dictionary, you will see the Item row. Type in the Item, eg. "R2C2". Close the application.



- To see if the link works, open InTouch Window Viewer, start FMS-DDE, start Excel. Make sure the Excel file is open to have the link.
- In the Excel file, add a number, eg. 10 in the cell R2C2 (Row 2 and Column 2).
- Name the file "TEXT1".



- Go the Window Viewer and see the link.



9 Using the PCS System as Single Stations

Besides using the system as a complete system which communicates via Profibus, the stations can be separated and worked on individually. When the stations are working individually, the communication to the controller is by RS 232.

9.1 Changing the Controller Dip Switches

- Remove the cover of the control section to be able to access the controller.
- At the back of the controller, remove the screws and pull out the controller interface card.
- To use it as a single station, the dip switches on the controller must be changed.
- The settings can be found in the controller manual.
- Set to RS 232 setting. (Off, Off, Off, Off, On, On, Off, Off)
- Reassemble the controller.
- Replace the cover of the control section.
- Connect the RS 232 cable to the controller and the other end to the computer's COM port. It can be either COM 1 or COM 2, depending on which is available. If you are using an older computer, you might need to get an adapter (9-pin to 25-pin).

9.2 Changing the Controller Menu Settings

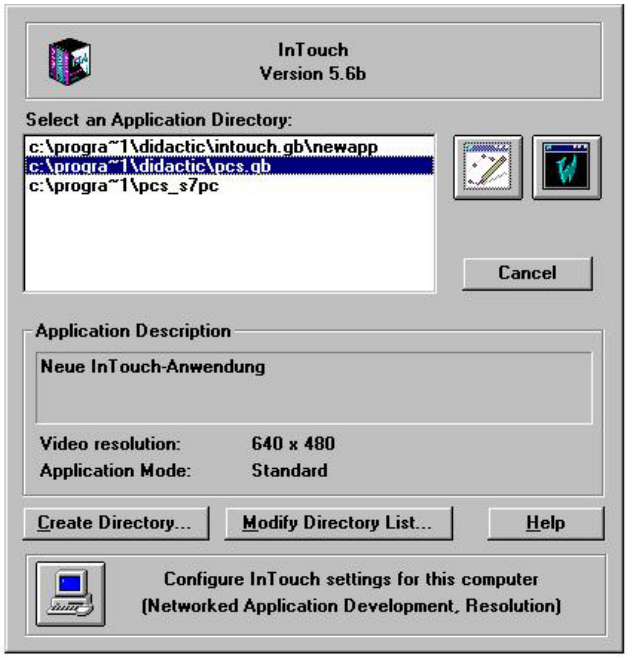
- Check the controller settings:
  - Make sure that you are in configuration mode. (Press [Select] + [Enter] for at least 5 seconds)
  - Go to the Menu Serial and set the following:
    - Set to Remote
    - Baud rate 9600
    - Even parity
    - Address No. 1.
- (If the controller was initially on Remote, first switch to Local)

9.3 Installing EzDDE

- This software is from Festo and is given in diskette form. Place the disk in the disk drive and install.
- This software is to ensure that the programs can communicate with each other.
- It is different from the DDE software from Softing, that is used for the complete system.

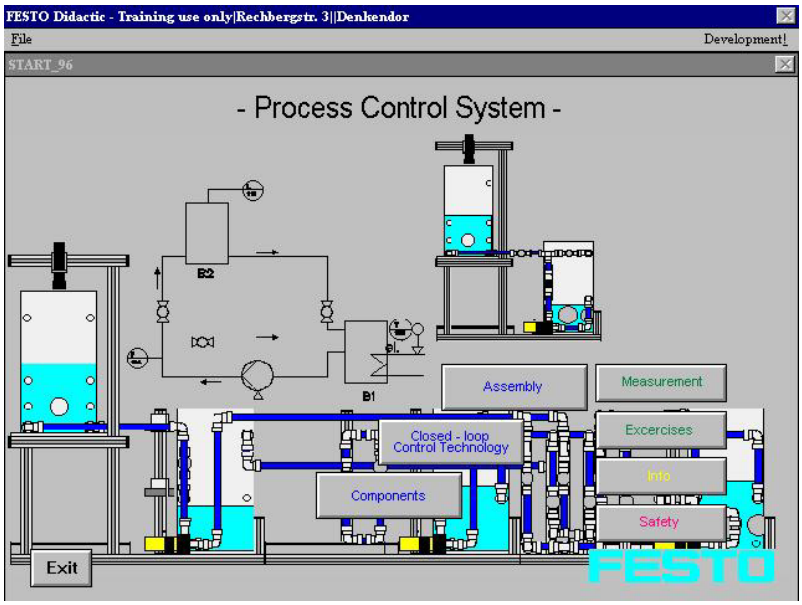
9.4 Installing InTouch Applications

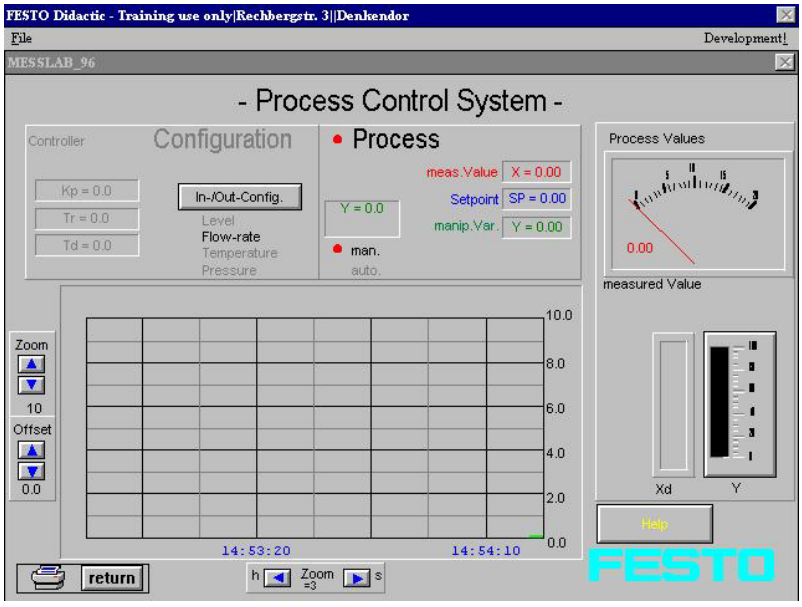
- Similar to installing the applications for the complete software, the files are installed in another directory and can be seen when you start InTouch.



9.5 Starting the Individual System

- Move on to your computer. Make sure the RS 232 cable is connected to your controller.
  - Start EzDDE. If it is not working, check : Festo Didactic - EzDDE - Config - Serial Driver - Edit - OK.
  - Start InTouch and select the appropriate application (eg. Flowrate).
- This is just to see if there is communication between the computer and the controller.





10 Courseware

As the PCS system is used for teaching PID control, the notes which are used by FESTO given to the participants. The documentation given are

- Application Foils 1
- Application Foils 2
- Set of Foils for Course
- Exercises
- The Saga of PID
- Participants Handout
- Workbook
- Course Description
- Course Objective

FESTO

P.O. Box 10 07 10
D-73707 Esslingen

Tel.: +49 - 711 3467 0

Fax: +49 - 711 3467 318